

OpenPI Comet

Scientific Reviewer

Strengths · Weaknesses · Actionable fixes (NOT a paper tour)

8+3 min

arXiv 2512.10071 · BEHAVIOR Challenge 2025 · smoke-run paper club

Strengths

Strengths

- ✓ Reusable end-to-end recipe
- ✓ Coverage ablation (pt1→pt50)
- ✓ Specified RFT (N=3, keep successes)
- ✓ Inference / data ablations published
- ✓ Honest oracle headroom (0.611)
- ✓ Code + HF checkpoints out

Weaknesses

Weaknesses

✗ Q-score undefined in paper

✗ Ablations mostly on easy turning_on_radio

✗ Post-challenge missing test Q

✗ Oracle easy to misread as a deployable policy

✗ No error bars / instance N variance

✗ No modular / hierarchical baseline

Ablation critique — Tables 3-4

Ablation critique — Tables 3-4 limited to turning_on_radio

Easy / robust bring-up task · extrapolation to hard suite unsupported

Probe task

turning_on_radio

Default SR 0.30

All Table 3-4 knobs

measured here only

What is missing

- Hard tasks (e.g. 48/49)
- Multi-task Q impact
- Modular baseline
- Error bars / seeds
- Planner/RL data isolation

Table 3 as published (radio only)

rate tasks are not shown.

Table 3: Ablation study of control mode, action horizon, input modality, and image resolution on the turning on radio task.

Settings	Control Mode	Action Horizon	Input Modality	Image Resolution	Success Rate
#1	Temporal Ensemble	50	RGB	Head:224	0.00
	Receding Temporal			Wrist:224	0.00
	Receding Horizon				0.25
#2	Receding Horizon	8	RGB	Head:224 Wrist:224	0.00
		16			0.10
		50			0.25
		32			0.30
#3	Receding Horizon	32	RGB+Depth Image	Head:224	0.20
			RGB+Point Cloud	Wrist:224	0.30
			RGB		0.30
#4	Receding Horizon	32	RGB	Head:224 Wrist:224	0.30
				Head:720 Wrist:480	0.60

Actionable fixes checklist

☐ Define Q-score formally + instance protocol

☐ Hard-task ablation panel (not only radio)

☐ Publish post-challenge test Q

☐ Report mean $\pm$ std over seeds / instances

☐ Publish RFT balance weights

☐ Add hierarchical / skill-chaining baseline

☐ Held-out RFT evaluation

☐ Turn oracle union into a checkpoint router

Strong recipe paper — strengthen claims with hard-task evidence and defined metrics.

Q&A · timing 8+3